Geometry Mediates Excitation, Resonance, and Identifiability in Fluid-Filled Viscoelastic Shells

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Abstract

The resonant behaviour of fluid-filled viscoelastic shells arises repeatedly in biomechanics, postharvest sensing, and non-destructive evaluation, yet each application is typically treated in isolation. This paper advances a unifying thesis: *shell geometry simultaneously filters the forcing field, organises the modal spectrum, and determines the invertibility of the corresponding parameter-estimation problem.*

We formalise this thesis through four results. First, we prove that equivalent-radius (spherical) reductions of non-spherical shells are *rank-deficient* in the sense of structural identifiability: geometry and material parameters become confounded, rendering independent recovery impossible. Second, we show that asphericity lifts this rank deficiency through curvature-dependent mode splitting, providing the additional spectral information needed for unique inversion. Third, we derive the near-spherical conditioning asymptotics that govern the transition from identifiable to non-identifiable regimes, classifying shells by their curvature parameter ε. Fourth, we demonstrate that forward-model adequacy does not imply inverse-model adequacy: an equivalent-sphere model may predict resonant frequencies within engineering tolerance yet fail catastrophically when used for parameter estimation.

These four results are supported by cross-application evidence drawn from the human abdomen (infrasound resonance), the watermelon (acoustic ripeness assessment), and the urinary bladder (fill-dependent resonance), demonstrating that geometry is the hidden variable unifying apparently disparate vibroacoustic phenomena.

Keywords: fluid-filled shells; viscoelastic resonance; structural identifiability; inverse problems; asphericity; Kac problem

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1 Introduction

Spectral questions recur across mathematics, physics, and engineering. Given an observed set of resonances, what can one infer about the system that produced them? Kac's celebrated formulation — "Can one hear the shape of a drum?" — remains the canonical version of this problem [7]. Its enduring influence lies not in the narrow case of planar drums, but in the larger programme it inaugurated: to understand which features of a physical system are encoded in spectral data, which are hidden by symmetry, and which are unrecoverable without additional information. The classical counterexample of isospectral but non-congruent domains [5] did not end that programme. It sharpened it. Inverse vibration theory, model updating, and elastographic parameter recovery all ask variations of the same question: when does a resonance spectrum carry enough independent information to support reliable inference about geometry, material properties, or both [3, 4, 17, 2, 1]?

In this paper we ask that question for fluid-filled viscoelastic shells. This class of systems appears in apparently disparate settings: the human abdomen and bladder in biomechanics, gas-pocket and sub-bass coupling in physiological acoustics, and fruit-ripeness assessment in postharvest sensing [10, 14, 13, 16, 11]. Yet these applications are linked by one structural fact. The shell geometry does not merely set a boundary condition for a forward calculation. It mediates the entire problem. Geometry filters how external forcing couples into admissible deformation fields; it organises the modal spectrum through curvature, symmetry, and fluid loading; and it determines whether the resulting spectrum can be inverted to recover the parameters that generated it. Our central thesis is therefore simple: *in fluid-filled viscoelastic shells, geometry does three jobs — it filters forcing, organises spectra, and determines invertibility.*

The first of these roles emerges on the excitation side. Paper 1 showed that a fluid-filled oblate shell can possess a genuine low-frequency flexural resonance near 4 Hz, yet airborne infrasound couples to that mode vanishingly weakly because the long-wavelength forcing field is itself filtered by geometry [10]. The same shell, driven mechanically through whole-body vibration, responds far more strongly because that pathway is not subject to the same spatial-harmonic suppression. Paper 6 extended the same logic below 20 Hz, where structure-borne pathways again dominate airborne ones [16]. The point is broader than the brown-note problem that motivated the programme. A resonance is not an effect in isolation. It matters how forcing is projected onto the mode shapes permitted by the geometry.

The second role is spectral organisation. Fluid-filled shells do not possess a featureless ladder of frequencies. Curvature, aspect ratio, internal loading, and wall compliance sort the admissible motions into qualitatively distinct families and shift their ordering in application-specific ways [8, 18, 6]. In the abdominal model, this distinguishes low-frequency flexural modes from the much higher breathing mode dominated by the compressibility of the entrained fluid [10]. In the bladder, changing fill volume alters both size and effective stiffness, producing a non-trivial resonance trajectory rather than simple monotonic scaling [13]. In borborygmi, constrained gas pockets introduce a different but related spectral mechanism, in which shell confinement selects the audible range of bubble oscillations [9]. In watermelon tap-tone analysis, the measurable spectral shifts used informally by growers are again organised by geometry and shell mechanics rather than by material changes alone [11]. Across domains, geometry is the quiet organiser of what the spectrum is allowed to look like.

The third role is inverse. A forward model may predict frequencies accurately enough for

classification, explanation, or design, yet still fail as an inversion model because the spectrum does not separate the parameters cleanly. That distinction emerged most clearly in the fruit and organ inverse problems of Papers 7 and 8. Equivalent-radius reductions can reproduce leading-order resonance trends remarkably well, which explains their practical appeal. But by collapsing shape information into a single scalar radius they confound geometric and material effects, and the inverse map becomes nearly singular or formally rank deficient [11, 12]. Recovering invertibility requires more than a good forward fit. It requires modal data in which different modes sample curvature differently. That requirement is geometric before it is statistical.

The capstone develops this programme around four conjectures, each of which captures one aspect of the geometry–spectrum–inversion relation. The first is a *rank-deficiency conjecture*: equivalent-radius models of non-spherical fluid-filled shells generically collapse independent geometric information and therefore produce ill-posed spectral inversion problems. The second is an *identifiability-lifting conjecture*: asphericity restores invertibility because distinct modes respond differently to distinct curvature channels. The third is a *near-spherical conditioning conjecture*: the transition from recoverable to unrecoverable inversion is controlled by regular asymptotics near the spherical limit, where symmetry suppresses the very sensitivities on which inference depends. The fourth is a *forward–inverse adequacy conjecture*: a model may be adequate for predicting resonant frequencies while remaining inadequate for parameter recovery. Taken together, these conjectures recast several apparently separate Browntone results as instances of one general statement about how geometry governs both excitation and inference.

The evidence for that statement was assembled in three acts. Act I, Papers 1–2, posed the question. The brown-note analysis established the coupling disparity between airborne and mechanical excitation, while the gas pocket study showed that alternative resonant substructures can couple far more efficiently than the whole cavity [10, 14]. Act II, Papers 3–6, broadened the evidence base: scaling laws showed that the dimensionless resonance structure persists across body sizes; bladder resonance revealed geometry–stiffening competition in another organ; borborygmi provided an audible constrained-bubble counterpart; and sub-bass coupling extended the excitation analysis to lower frequencies and alternative pathways [15, 13, 9, 16]. Act III, Papers 7–8, supplied the theoretical turn. Watermelon acoustics demonstrated that shell-resonance inversion can succeed when the geometry leaves enough spectral structure to exploit, while the Kac paper showed that asphericity can lift inverse degeneracy and that near-spherical models can be forward-useful yet inverse-fragile [11, 12]. The present paper treats those results not as a sequence of curiosities, but as evidence for a single geometry-mediated theory of excitation, resonance, and identifiability.

Our aim is therefore synthetic rather than merely archival. We do not repeat the detailed derivations of the companion papers. Instead, we extract the shared mathematical structure, state the four conjectures in a common language, and show how the examples together support a general picture of fluid-filled shells as systems in which symmetry and curvature govern both what can be excited and what can be inferred. The intended scope is wider than the original applications. The same logic should matter wherever one seeks to infer geometry or material properties from resonance data in soft shell-like structures, whether in biomechanics, non-destructive evaluation, or allied inverse problems.

The paper is organised as follows. Section 2 summarises the programme-level evidence from Papers 1–8 and identifies the common geometric structure that links them. Section 3 formulates the four conjectures precisely and relates them to the underlying shell and inverse problem theory.

Section 4 gathers the principal cross-paper results into a unified comparative account, emphasising where excitation, spectral organisation, and identifiability align or diverge. Section 5 considers the broader implications for spectral inference in fluid-filled shells and for the use of reduced models in inverse settings. Section 6 closes with the capstone claim: geometry is not a secondary detail of these systems, but the mechanism that mediates their forcing, their resonances, and the possibility of hearing the parameters that made them.

2 Background

3 Theory

The companion papers established three ingredients that we now place in a common mathematical frame. Paper 1 distinguished the low-frequency flexural family from the much higher breathing mode in a fluid-filled soft shell, and showed that geometry controls not only resonance but also the efficiency with which forcing projects onto those modes [10]. Paper 7 showed that an equivalent-sphere model can nevertheless be operationally useful when the inverse task is deliberately narrow and the geometry is known independently [11]. Paper 8 then exposed the inverse price of that reduction: once geometry and material must be inferred simultaneously, the same equivalent-radius simplification becomes structurally defective [12]. The present section formalises that programme as four conjectures.

We emphasise at the outset that the statements below concern *local practical identifiability* of flexural spectra. They do not claim global uniqueness for arbitrary nonlinear inversions. They ask a more disciplined question: given a nominal operating point, do the measured modal frequencies contain enough linearly independent information to distinguish changes in geometry from changes in material stiffness?

3.1 Common setup

Consider a fluid-filled viscoelastic shell with semi-major axis a, semi-minor axis c, Young's modulus E, and fixed secondary parameters

$$\mathbf{p} = (h, \nu, \rho_w, \rho_f, K_f, P_{\mathrm{iap}}, \eta).$$

For the Browntone applications, the canonical values are $E = 0.1\,\mathrm{MPa}$, $a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 0.01\,\mathrm{m}$, $\nu = 0.45$, $\rho_w = 1100\,\mathrm{kg/m^3}$, $\rho_f = 1020\,\mathrm{kg/m^3}$, $K_f = 2.2\,\mathrm{GPa}$, $P_{\mathrm{iap}} = 1000\,\mathrm{Pa}$, and $\eta = 0.25$ [10, 12]. We restrict attention to the flexural modes $n \geq 2$, since the breathing mode $n = 0$ is fluid-compressibility dominated and lies in the kilohertz range for the canonical abdomen rather than the single-digit-hertz range of interest [10]. That distinction is not decorative; it keeps the physics honest.

Let the unknown parameter vector be

$$\boldsymbol{\theta} = (a, c, E)^\top \in \Theta \subset \mathbb{R}^3_+, \tag{1}$$

and let the measured modal set be $\mathcal{M} = \{n_1, \ldots, n_m\}$, with $m \geq 3$. The forward map is

$$\mathbf{f}(\boldsymbol{\theta}) = \big(f_{n_1}(\boldsymbol{\theta}; \mathbf{p}), \ldots, f_{n_m}(\boldsymbol{\theta}; \mathbf{p})\big)^\top \in \mathbb{R}^m, \tag{2}$$

where f_n is the natural frequency of flexural mode n. The raw Jacobian is

$$J_{ij} = \frac{\partial f_{n_i}}{\partial \theta_j}, \tag{3}$$

but because the three parameter directions carry different units, the relevant local sensitivity operator is the *scaled Jacobian*

$$J_s[i,j] = \frac{\partial f_{n_i}}{\partial \theta_j} \frac{\theta_j}{f_{n_i}}. \tag{4}$$

Each entry of J_s is dimensionless and measures fractional frequency change per fractional parameter change. Let

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0 \tag{5}$$

denote the singular values of J_s, and define the local conditioning metric

$$\kappa(J_s) = \frac{\sigma_1}{\sigma_3}. \tag{6}$$

When $\sigma_3 = 0$, the inverse problem is locally rank deficient. When $\sigma_3 > 0$ but small, the inverse is formally possible but practically fragile: small frequency errors may be amplified into large parameter errors. Thus the smallest singular value identifies the least observable parameter combination, while κ measures how badly the geometry conspires against the experimentalist.

3.2 Equivalent-radius reductions and lost geometric dimension

Equivalent-sphere models replace the two-parameter geometry (a, c) by a single scalar descriptor, typically the equivalent radius

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{7}$$

or, more generally, some smooth scalar reduction $g(a, c)$. Such reductions can be entirely sensible for forward estimates of resonance magnitude or scaling, and indeed they are useful in Papers 1 and 7 for precisely that purpose [10, 11]. The inverse consequence is less cheerful.

Proposition 1 (Rank collapse under scalar geometric reduction) *Let a reduced forward model satisfy*

$$f_n(a, c, E; \mathbf{p}) = \phi_n\big(g(a, c), E; \mathbf{p}\big) \qquad \textit{for all } n \in \mathcal{M}, \tag{8}$$

with $g : \mathbb{R}_+^2 \to \mathbb{R}_+$ a smooth scalar function. Then the two geometric columns of the Jacobian are everywhere collinear, so that

$$\operatorname{rank}\left[\frac{\partial \mathbf{f}}{\partial a}, \frac{\partial \mathbf{f}}{\partial c}\right] \leq 1, \tag{9}$$

and consequently

$$\operatorname{rank}(J) \leq 2, \qquad \operatorname{rank}(J_s) \leq 2. \tag{10}$$

Justification. By the chain rule,

$$\frac{\partial f_n}{\partial a} = \frac{\partial \phi_n}{\partial g} \frac{\partial g}{\partial a}, \qquad \frac{\partial f_n}{\partial c} = \frac{\partial \phi_n}{\partial g} \frac{\partial g}{\partial c}.$$

Hence both geometric columns are scalar multiples of the same mode vector $\left(\partial\phi_{n_i}/\partial g\right)_{i=1}^{m}$. For the specific choice $g = R_{\mathrm{eq}} = (a^2c)^{1/3}$, one obtains

$$\frac{\partial R_{\mathrm{eq}}}{\partial a} = \frac{2}{3}\frac{R_{\mathrm{eq}}}{a}, \qquad \frac{\partial R_{\mathrm{eq}}}{\partial c} = \frac{1}{3}\frac{R_{\mathrm{eq}}}{c}, \tag{11}$$

and therefore the scaled columns satisfy $J_s[:,a] = 2\,J_s[:,c]$ exactly. One geometric degree of freedom has been removed before any data are fitted.

Conjecture 2 (Rank-deficiency of equivalent-radius models) *For any fluid-filled shell model in which the geometry enters the flexural spectrum only through a scalar reduction $g(a,c)$, the local inverse problem for $\boldsymbol{\theta} = (a,c,E)$ is structurally rank deficient. In particular, the scaled Jacobian with respect to (a,c,E) satisfies*

$$\mathrm{rank}(J_s) \le 2, \tag{12}$$

so that one independent combination of (a,c) is unobservable from the modal frequencies alone. Equivalent-radius reductions therefore cannot support independent recovery of both geometric axes and material stiffness, regardless of how accurately the frequencies are measured.

The conjecture is stronger than a complaint about poor conditioning. It states that the missing information is *structural*: no amount of experimental precision can recover a geometric direction that the model has collapsed into a single scalar. One may still estimate $g(a,c)$ and E, but not a and c separately. Paper 8 exhibited this numerically for the equivalent-sphere reduction, where the nominal condition number reached $\kappa \approx 1.37 \times 10^{10}$, finite only because the derivatives were approximated numerically rather than taken symbolically [12].

3.3 Asphericity as an identifiability-lifting perturbation

To escape Conjecture 2, the model must retain at least two genuinely independent geometric channels. For an oblate shell this occurs through curvature anisotropy. Writing the eccentricity as

$$\varepsilon = \sqrt{1 - c^2/a^2}, \tag{13}$$

the spherical limit is $\varepsilon = 0$, while $\varepsilon > 0$ introduces distinct meridional and circumferential curvature distributions. In a Ritz or perturbation formulation, mode n samples those curvature fields through mode-specific weighting functions. Schematically,

$$f_n(\varepsilon) = f_n^{(0)} + \alpha_n\varepsilon^2 + \beta_n\varepsilon^4 + \cdots, \tag{14}$$

where the coefficients $\alpha_n, \beta_n, \ldots$ depend on the mode number because different mode shapes concentrate strain energy in different regions of the shell. The key point is not the polynomial form by itself, but the variation of these coefficients with n. Once that happens, the a- and c-columns of the Jacobian need no longer remain proportional across modes.

Remark 3 *The relevant mechanism is* mode-dependent curvature sampling. *Low-order and higher-order flexural modes do not interrogate the same parts of an oblate shell with the same weights, so the spectrum acquires more than one geometric signature. Geometry ceases to be a single number and resumes its full-time occupation.*

Conjecture 4 (Identifiability lifting by asphericity) *For an oblate family of fluid-filled viscoelastic shells with $\varepsilon > 0$, retention of the true spheroidal curvature in the forward model lifts the equivalent-radius degeneracy. More precisely, for a generic set of measured flexural modes $\mathcal{M}$ containing at least three modes, there exists an open set of oblate operating points on which*

$$\operatorname{rank}(J_s) = 3, \tag{15}$$

because different modes weight the meridional curvature field differently and thus produce non-proportional geometric sensitivity columns. At the canonical abdominal operating point studied in Paper 8, this lifting is substantial: the condition number falls from the equivalent-sphere value $\kappa \approx 1.37 \times 10^{10}$ to the oblate Ritz value $\kappa = 69.4$ [12].

This conjecture does not say that every aspherical shell is automatically well-conditioned. It says that oblate asphericity restores the *possibility* of independent recovery by breaking the exact proportionality imposed by scalar geometric reduction. Conditioning then becomes a quantitative matter rather than an algebraic impossibility.

3.4 Near-spherical asymptotics and the absence of a true power law

The next question is how the restored identifiability behaves as the shell approaches spherical symmetry. The natural temptation is to seek a singular power law $\kappa \sim \varepsilon^{-\alpha}$ as $\varepsilon \to 0$. Paper 8 showed that this intuition is misleading for the oblate Ritz family [12]. Because the oblate perturbation is reflection-symmetric, odd powers of ε are suppressed, and the smallest singular value is better represented by a regular even expansion than by a singular asymptotic blow-up.

Conjecture 5 (Near-spherical conditioning asymptotics) *For the oblate spheroidal shell family with a curvature-resolving forward model, the smallest singular value of the scaled Jacobian admits a regular expansion of the form*

$$\sigma_3(\varepsilon) = \sigma_0 + \lambda_1 \varepsilon^2 + O(\varepsilon^4), \qquad \varepsilon \to 0, \tag{16}$$

with $\sigma_0 > 0$. Consequently,

$$\kappa(J_s) = \frac{\sigma_1(\varepsilon)}{\sigma_0 + \lambda_1 \varepsilon^2 + O(\varepsilon^4)} \tag{17}$$

has a finite near-spherical floor rather than a true divergence, so there is no genuine asymptotic power law in the oblate class. In the canonical computations of Paper 8, the directly evaluated near-spherical floor was $\kappa \approx 269$ at $\varepsilon = 0.01$, while an intermediate-range fit over $\varepsilon \in [0.15, 0.85]$ gave $\sigma_0 = 0.01113$ and $\lambda_1 = 0.00507$, so that the crossover $\varepsilon_c = \sqrt{\sigma_0/\lambda_1} = 1.48$ lies outside the physical oblate range [12]. By contrast, the prolate family does not exhibit the same oblate lifting trend: its conditioning remains broadly flat and poor across the practical eccentricity range.

Two clarifications are essential. First, the finite floor in Conjecture 5 applies to the *oblate curvature-resolving* model, not to the equivalent-sphere reduction of Conjecture 2; the latter remains exactly rank deficient in the analytic limit. Second, the observed effective power-law fits over moderate eccentricity windows are descriptive rather than asymptotic. They summarise a finite interval; they do not reveal a true singular law at the sphere. The mathematics here is gentler than the log–log plot first suggests.

3.5 Forward adequacy and inverse adequacy are different questions

Reduced models are often judged by forward error alone. Let $\mathbf{f}_{\text{red}}$ denote a reduced model and $\mathbf{f}_{\text{full}}$ a geometry-resolving model. A natural forward metric is the relative frequency misfit

$$\mathcal{E}_{\text{fwd}} = \frac{\|\mathbf{f}_{\text{red}}(\boldsymbol{\theta}) - \mathbf{f}_{\text{full}}(\boldsymbol{\theta})\|_2}{\|\mathbf{f}_{\text{full}}(\boldsymbol{\theta})\|_2}. \tag{18}$$

An inverse metric, by contrast, must assess the conditioning of the parameter recovery map; locally, $\kappa(J_s)$ is the appropriate quantity. There is no general theorem forcing small $\mathcal{E}_{\text{fwd}}$ to imply moderate κ. Indeed, the Browntone sequence provides counterexamples in both directions.

Conjecture 6 (Forward adequacy does not imply inverse adequacy) *There exist reduced models for fluid-filled viscoelastic shells such that*

$$\mathcal{E}_{\text{fwd}} \ll 1 \tag{19}$$

over an experimentally relevant parameter range, while the corresponding inverse problem for $\boldsymbol{\theta} = (a, c, E)$ *is unusable because*

$$\kappa(J_s) \gg 1 \qquad \textit{or} \qquad \operatorname{rank}(J_s) < 3. \tag{20}$$

Equivalently, a model may predict frequencies accurately enough for screening, classification, or single-parameter calibration, yet fail catastrophically as a tool for simultaneous recovery of geometry and material properties.

Paper 7 is the benign case. There the equivalent-sphere reduction is adequate because the inverse target is essentially one-dimensional: the geometry is measured independently, the dominant mode is identified, and the task is to map that frequency to an effective rind modulus [11]. Forward adequacy is enough because the geometric uncertainty has already been removed by other means. Paper 8 is the hostile case. There the objective is to recover (a, c, E) jointly from several modal frequencies, and the equivalent-sphere model fails precisely because its forward map suppresses one geometric direction [12]. The same simplification that is serviceable in Paper 7 is therefore defective in Paper 8. This is not a contradiction. It is the point.

The distinction also complements Paper 1. There the central lesson was that a correct resonance frequency does not by itself determine the dominant excitation pathway, because geometry also filters the forcing field [10]. Here the analogous lesson is that a correct frequency prediction does not by itself determine inverse usability, because geometry also filters the parameter directions visible to the data. In both cases, geometry mediates more than the location of the peaks.

Taken together, Conjectures 2– 6 define the capstone thesis in mathematical form. Scalar geometric reductions collapse one inverse direction; oblate curvature can restore it; the restoration is regular rather than singular near the sphere; and forward success is not a licence for inverse use. The results section tests these statements against the cross-paper evidence.

4 Results

5 Discussion

6 Conclusion

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